Course 02323/02402 Introduction to Statistics

Lecture 9: Multiple linear regression

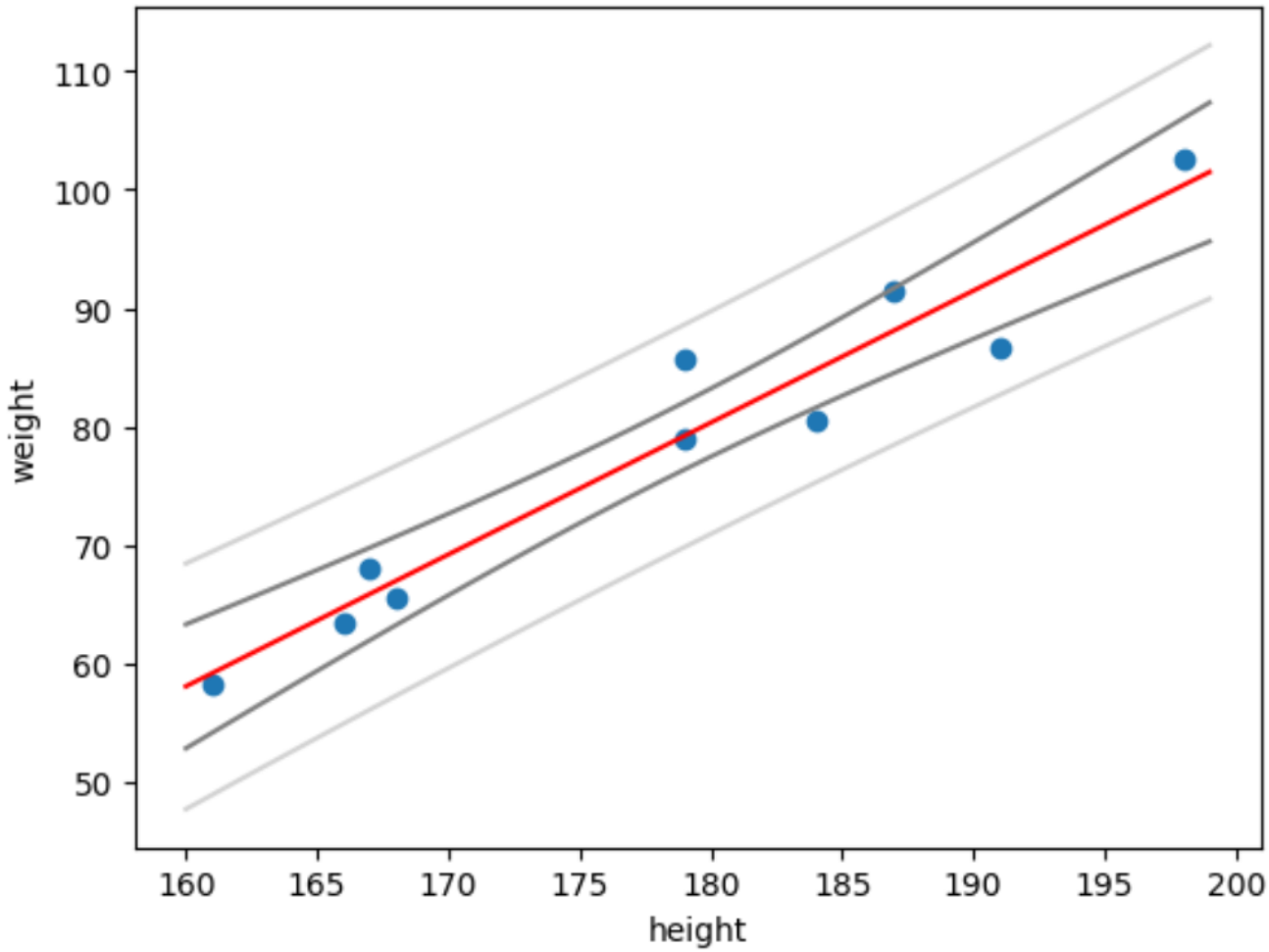
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Agenda

- Summary
- Motivational Example: Ozon concentration

1. Multiple linear regression
   - Mindste kvadraters metode (Least squares)
   - Hypothesis test and confidence interval for the β_i's
   - Confidence and prediction intervals
2. Model control (residual analysis)
3. Model Development
   - Curvilinearity
   - Collinearity
   - Model selection

Summary: Simple Linear Regression



Example: Ozon concentration

- Go to today's Python notebook in VS Code
  - "Example: Ozon concentration"
  - Kahoot x4



Overview

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Multiple linear regression

- The *linear regression model* (general linear model, GLM):

$$Y_i = \beta_0 + \beta_1 x_{1,i} + \ldots + \beta_p x_{p,i} + \varepsilon_i \quad (i = 1, \ldots, n).$$

- Y_i is the *dependent variable*.
- $x_{1i}, x_{2i}, \ldots, x_{pi}$ are *explanatory variables*.
- There are a total of p explanatory variables in the model.
- ε_i is the deviation (residual).
- We assume $\varepsilon_i \sim N(0, \sigma^2)$ (and i.i.d.).

Matrix notation

The *linear regression model* (general linear model, GLM):

$$Y_i = \beta_0 + \beta_1 \cdot x_{1i} + \beta_2 \cdot x_{2i} + \ldots + \varepsilon_i$$

In matrix form:

$$\mathbf{Y} = \mathbf{X}\beta + \varepsilon$$

$$\mathbf{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} \qquad \mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{21} & \ldots \\ 1 & x_{12} & x_{22} & \ldots \\ \vdots & \vdots & \vdots & \vdots \\ 1 & x_{1n} & x_{2n} & \ldots \end{bmatrix} \qquad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \end{bmatrix} \qquad \varepsilon = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}$$

Example: Ozone concentration

- Go to today's Python notebook in VS Code
  - "Example: Ozone concentration with Multiple Linear Regression"



Least Squares Method (*Least Squares*)

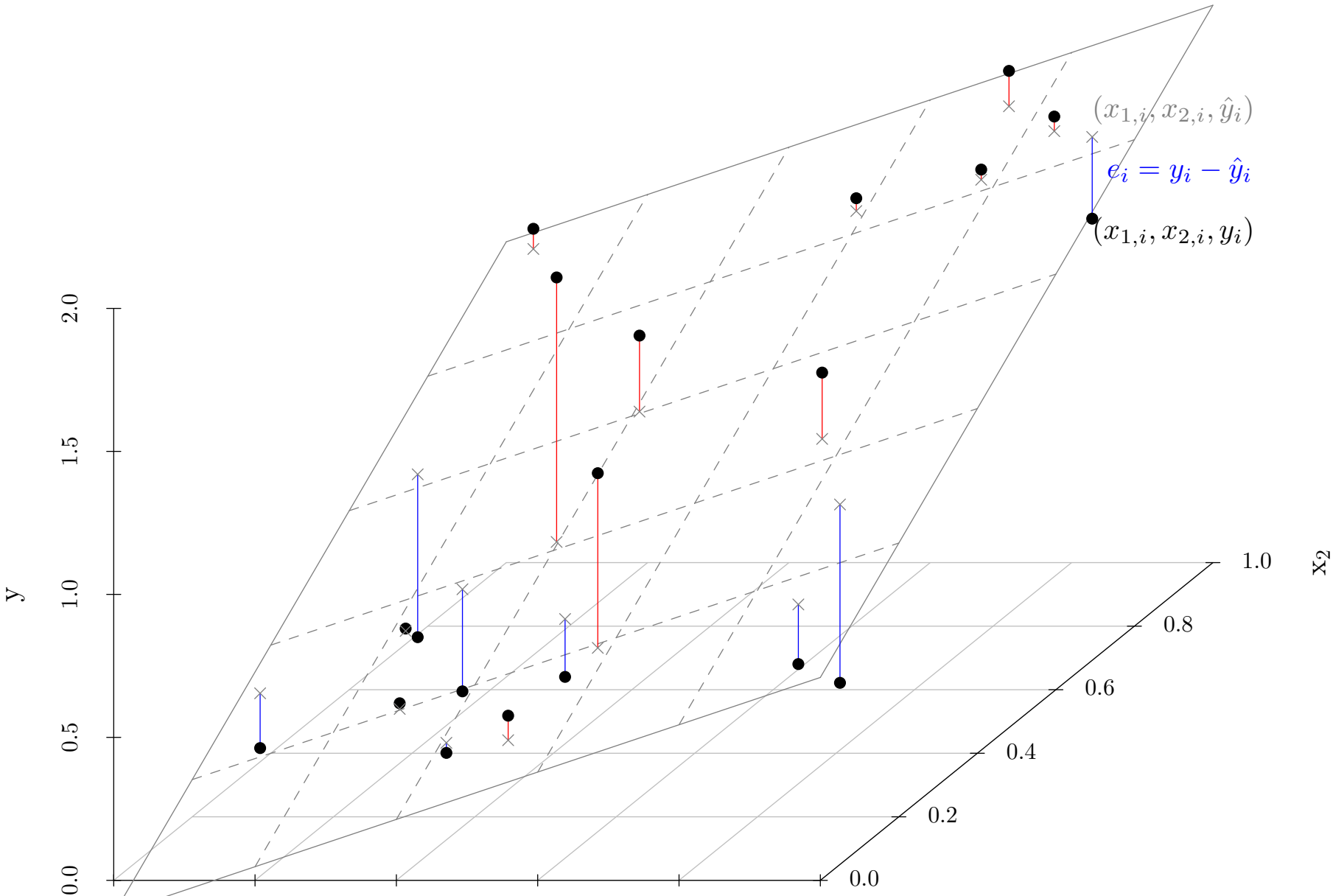
- We want to **estimate** the parameters β_0, β_1, ... ,β_p.
- A good idea: Let us minimize the variance of the residuals (σ^2).
- We minimize the sum of squared residuals ("Residual Sum of Squares", RSS):

$$RSS = \sum_{i=1}^{n} \varepsilon_i^2 = \varepsilon^T \varepsilon$$

- $\varepsilon_i^2 = (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{1,i} - \hat{\beta}_2 x_{2,i} - \ldots - \hat{\beta}_p x_{p,i})^2 = (y_i - \hat{y}_i)^2$
- ε is a *vector* containing all residuals

We minimize RSS with respect to all $(p+1)$ β's.

Least squares method



Python

Use Python to estimate a linear regression model:

- `my_fit = smf.ols(formula = 'y ~ x1 + x2 + x3', data=...).fit()`
- Note: "`y ~ x1 + x2 + x3`"

Print the regression table:

- `print(my_fit.summary(slim=True))`

- Go to Python notebook: "week9_python.ipynb" i VS Code
  Example: Ozon concentration with multiple linear regression



Least Squares Estimators

Theorem 6.17

The estimators of the parameters in the simple linear regression model are given by

$$\hat{\boldsymbol{\beta}} = (\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{Y}, \tag{6-44}$$

and the covariance matrix of the estimates is

$$V[\hat{\boldsymbol{\beta}}] = \sigma^2(\boldsymbol{X}^T\boldsymbol{X})^{-1}, \tag{6-45}$$

and central estimate for the residual variance is

$$\hat{\sigma}^2 = \frac{RSS}{n-(p+1)}. \tag{6-46}$$

Note: For *multiple* linear regression, the formulas for $\hat{\beta}_i$ og $\hat{\sigma}_{\beta_i}$ are given *only* in matrix form (Python performs the calculations for us).

Hypothesis test and confidence interval for the β_i's

Theorem 6.2 Hypothesis tests and confidence intervals

Suppose the we are given parameter estimates $(\hat{\beta}_0, \ldots, \hat{\beta}_p)$ and their corresponding standard errors $(\hat{\sigma}_{\beta_0}, \ldots, \hat{\sigma}_{\beta_p})$, then under the null hypothesis

$$H_{0,i}: \quad \beta_i = \beta_{0,i}, \tag{6-15}$$

the t-statistic

$$T_i = \frac{\hat{\beta}_i - \beta_{0,i}}{\hat{\sigma}_{\beta_i}}, \tag{6-16}$$

will follow the t-distribution with $n - (p+1)$ degrees of freedom, and hypothesis testing and confidence intervals should be based on this distribution. Further, a central estimate for the residual variance is

$$\hat{\sigma}^2 = \frac{RSS(\hat{\beta}_0, \ldots, \hat{\beta}_p)}{n - (p+1)}. \tag{6-17}$$

Confidence interval for β_0 and β_1

Method 6.5 Parameter confidence intervals

$(1-\alpha)$ confidence interval for β_i is given by

$$\hat{\beta}_i \pm t_{1-\alpha/2}\,\hat{\sigma}_{\beta_i}, \tag{6-20}$$

where $t_{1-\alpha/2}$ is the $(1-\alpha/2)$-quantile of a t-distribution with $n-(p+1)$ degrees of freedom.

Remember: $\hat{\beta}_i$ og $\hat{\sigma}_{\beta_i}$ are calculated using the formulas in matrix form *or* with Python

Kahoot!

(x2)

Thinking Break

$$\hat{\beta}_{wind} \pm t_{1-\alpha/2} \cdot \hat{\sigma}_{\beta_{wind}}$$

$$-0.0693 \pm t_{1-\alpha/2} \cdot 0.015 \quad \text{(from the example in Python)}$$

(For the 95% confidence interval, use $\alpha = 0.05$)

What is $t_{1-\alpha/2}$ (from which distribution)?

_____

_____

_____ (also draw the distribution and mark $t_{1-\alpha/2}$)

How would we find $t_{1-\alpha/2} = t_{0.975}$ in Python (write the code)?

_____

What is then the 90% confidence interval for β_{wind}?

_____

Hypothesis test for the β_i's

Method 6.4 Level α t-tests for parameters

1. Formulate the *null hypothesis*: $H_{0,i} : \beta_i = \beta_{0,i}$, and the alternative hypothesis $H_{1,i} : \beta_i \neq \beta_{0,i}$
2. Compute the test statistic $t_{\text{obs},\beta_i} = \frac{\hat{\beta}_i - \beta_{0,i}}{\hat{\sigma}_{\beta_i}}$
3. Compute the evidence against the *null hypothesis*

$$p\text{-value}_i = 2P(T > |t_{\text{obs},\beta_i}|) \tag{6-19}$$

4. If the p-value$_i < \alpha$ reject $H_{0,i}$, otherwise accept $H_{0,i}$

Kahoot!

(x2)

Interpretation of parameters (Remark 6.14)

What does $\hat{\beta}_i$ represent?

- The expected change in y when x_i changes by one unit.

Interpretation of parameters (Remark 6.14)

What does $\hat{\beta}_i$ represent?

- The expected change in y when x_i changes by one unit.
- The effect of x_i given the other variables.
- The effect of x_i adjusted for the effects of the other variables.
- The effect of x_i "when the other variables are unchanged".

Interpretation of parameters (Remark 6.14)

What does $\hat{\beta}_i$ represent?

- The expected change in y when x_i changes by one unit.
- The effect of x_i given the other variables.
- The effect of x_i adjusted for the effects of the other variables.
- The effect of x_i "when the other variables are unchanged".
- Depends on what else is in the model!
- In general: NOT a causal effect/intervention effect!

Confidence and Prediction Intervals

Method 6.9 Intervals for the line (by Python)

The (1-α) **confidence and prediction intervals** for the line $\hat{\beta}_0 + \hat{\beta}_1 x_{1,\text{new}} + \cdots + \hat{\beta}_p x_{p,\text{new}}$ are calculated in Python by

```
# Confidence and Prediction interval
fit.get_prediction(new_data).summary_frame(alpha=0.05)
```

Remark 6.10

Explicit formulas for confidence and prediction intervals are given in Section 6.6.

Confidence and prediction intervals - Method 6.9:

Calculate confidence interval and prediction interval with Python:

- `x_new = pd.DataFrame('x1':..., 'x2':..., 'x3':...)`
- `my_fit.get_prediction(x_new).summary_frame(alpha=...)`

- Go to today's Python notebook in VS Code
  - "Example: Confidence- and prediction interval"



Visual Studio Code

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Model control (residual analysis)

- Model control: Analyze the residuals to check if the assumptions are met.
- Same assumptions as for the simple linear model

$$Y_i = \beta_0 + \beta_1 x_{1,i} + \ldots + \beta_p x_{p,i} + \varepsilon_i \quad (i = 1, \ldots, n).$$

- We assume $\varepsilon_i \sim N(0, \sigma^2)$ og i.i.d.

Example: Ozone concentration

You can obtain residuals and fitted y-values directly from the "fit" variable:

- `my_fit = smf.ols(formula = 'y ~ x1 + x2 + x3', data=...).fit()`
- `residuals = my_fit.resid`
- `fittedvalues = my_fit.fittedvalues`

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  - "Example: Model control"



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A curvilinear model

Regression models for non-linear data based on Taylor expansions.

If we want to use a model of the type

$$Y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \varepsilon_i,$$

then we can use a multiple linear regression model

$$Y_i = \beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2} + \varepsilon_i,$$

where

$$x_{i,1} = x_i, \quad x_{i,2} = x_i^2$$

and use the same methods as for multiple linear regression.

Example: Ozone concentration

- Go to today's Python notebook in VS Code
  - "Example: curvilinear regression"
    (+ KAHOOT x1)



Be careful with predictions outside the data basis

Remark 6.12

In general one should be careful when extrapolation models into areas where there is no data, and this is in particular true when we use curvilinear regression.

Collinearity

- If two (or more) explanatory variables have a perfect linear relationship, then we cannot determine which is explanatory.
- Also an issue if the relationship is close to linear.
- Related to the concept of "confounders".
- With two highly correlated x-variables:
  - *Together*, neither may have a "unique" effect.
  - *Separately*, they may have a large effect.

Example: Ozone concentration

- Go to today's Python notebook in VS Code
  - "Example: Collinearity and confounding"
    (+ KAHOOT x1)



Interpretation of the β_i's

Remark 6.14 Interpretation of parameters

In general we can interpret the parameters of a multiple linear regression model as the effect of the variable given the other variables. E.g. β_j is the effect of x_j when we have accounted for other effects (x_i, $i \neq j$). This interpretation is however problematic when we have strong collinearity, because the true effects are hidden by the correlation.

Example: Ozone concentration

- Go to today's Python notebook in VS Code
  - "Example: Model selection"



Model extension (forward selection)

- *Not included in the book*
- Start with a *simple linear regression model* with one significant explanatory variable
- *Extend the model* with other explanatory variables one at a time
- *Stop* when there are no more significant extensions

Model reduction (backward selection)

- *Described in the book under section 6.5*
- Start with the full model
- Remove the "least significant" variable
- Stop when all remaining parameters are significant

Kahoot!

(x1)

Model selection

- There is no certain method to find the best model!
- Selecting a model requires subjective decisions.
- Different procedures, either forward or backward selection (or both), depend on the circumstances.
- Statistical methods and tests exist to compare models.
- In this course, only backward selection is described.

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